Salient Object Detection Based on Progressively Supervised Learning for Remote Sensing Images

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***Abstract*— Salient object detection (SOD) is a crucial task in the field of remote sensing image (RSI) processing. Weakly supervised SOD methods, which generate saliency maps by classification convolutional neural networks (CNNs), considerably reduce labor costs. However, due to the complexity of remote sensing scenes, concerns remain about weakly supervised SOD for RSIs: 1) since the pooling operations are applied in the classification CNNs, the boundary maintenance of weakly supervised methods is unsatisfactory and 2) several sophisticated postprocessing procedures are used in previous weakly supervised methods, which are inevitably time-consuming. To solve these problems, we combine the benefits of weakly and fully supervised learning and propose a new SOD method named progressively supervised learning (PSL) for RSIs. The proposed method realizes end-to-end SOD with a lightweight model under imagewise annotations. First, to reduce the demands on large-scale pixelwise annotations, we propose a pseudo-label generation method based on a classification network and gradient-weighted class activation mapping (Grad-CAM) to compute pseudo saliency maps (PSMs) for training samples and auxiliary images in a weakly supervised manner. Then, to improve the computational efficiency, we construct a feedback saliency analysis network (FSAN), where the generated PSMs are regarded as pixelwise labels. Finally, inspired by curriculum learning, we design a new denoising loss function to further reduce the effect brought by missing judgment in PSMs and enhance the detection accuracy. Comprehensive evaluations with two remote sensing data sets and a comparison with 11 methods validate the superiority of the proposed PSL model.**

***Index Terms*— Convolutional neural network (CNN), fully supervised learning, remote sensing image (RSI) processing, salient object detection (SOD), weakly supervised learning (WSL).**

I. Introduction

SALIENT object detection (SOD) aims at locating objects that are most visually attractive and distinctive in an image [1], [2]. Previous studies have demonstrated the remarkable performance of SOD in natural scenes [3]–[5], especially after the rise of deep learning (DL) technology [1], [5], [6]. In this article, we focus on SOD under low label costs with a lightweight model and high accuracy for remote sensing images (RSIs).

It is worth noting that targets in RSIs always exhibit three characteristics.

1) The targets always consist of several closed areas, and their shape and position have strong randomicity.
2) There is a low spectral similarity of adjacent pixels inside the target. In addition, the edges of the targets are elaborate.
3) Under the same illumination and scale conditions, one or more features of targets, such as brightness, spectrum, texture, edge, and local contrast, are more salient in visual characteristics.

Although there are discrepancies in the shape and spectral characteristics of targets in RSIs, humans can quickly focus on discriminative targets [7]–[9]. Considering the strong cognitive abilities of human beings [10], [11], researchers have brought SOD into the field of RSI interpretation, which can help to extract regions of interest (ROIs) automatically by simulating the human visual system [2], [12], [13]. Over the last two decades, unsupervised methods [9], [14]–[16] and supervised methods have become the two mainstream approaches.

Unsupervised methods tend to design multiscale man-made features and then compute the final result by a threshold or classifier [9], [14]–[19]. In previous work, the quaternion Fourier transform (QFT) [9], multiscale spectrum residual method [17], scale-invariant feature transform [19], [20], and support vector machine [19], [21] are introduced for ROI extraction and SOD in RSIs. However, the visual characteristics of RSIs generated from different satellites vary widely. Man-made features often fail to provide a satisfactory performance.

Convolutional neural networks (CNNs) [22]–[24], which are the most popular supervised methods, have delivered record-breaking performances in many vision tasks, such as image classification [23]–[25], image segmentation [26], and superresolution reconstruction [27], [28]. For SOD, previous studies have shown that CNNs are very useful for handling complex scenes. At the same time, since fully convolutional neural networks (FCNs) [29] can take arbitrary size input and overcome problems such as blurring and inaccurate predictions [30], they have been widely introduced into SOD for RSIs [31]–[35].

Earlier works in the field of SOD tend to learn the mapping from the image space to the output saliency maps under pixelwise annotations, which we also call them fully supervised methods. As for this kind of methods, researchers

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mostly focus on the structures of neural networks and achieve many satisfactory results. Kampffmeyer *et al.* [31] developed patch-based and pixel-to-pixel-based CNN architectures for the segmentation of urban RSIs to map land cover with small objects. Chen *et al.* [32] designed symmetrical dense-shortcut deep FCNs for the semantic segmentation of very-high-resolution RSIs. Diao *et al.* [33] combined the strength of the location regression-based CNN framework and the salient features for target detection in SAR images. Shao *et al.* [34] proposed a saliency-aware CNN framework for ship detection, comprising comprehensive ship discriminative features such as deep features, saliency maps, and coastline priors.

Although the abovementioned fully supervised methods can achieve high performance, all the positive training images should be manually labeled with pixelwise annotations. This is incredibly laborious, and the dependence on manual annotations would restrict the scalability. To this end, weakly supervised methods, which use only imagewise annotations, have been extensively studied because they do not require expensive human effort. The most representative methods include the computation of gradient maps [36], global max-pooling (GMP) method [37], class activation maps (CAM) method [38], gradient-weighted class activation mapping (Grad-CAM) [39], and Grad-CAM++ [40].

For RSIs, Yao *et al.* [41] proposed a weakly supervised learning (WSL) framework based on coupled CNNs for aircraft detection. Zhang *et al.* [42] proposed a unified annotation framework by combining discriminative high-level feature learning and weakly supervised feature transferring. Singh and Komodakis [43] built a classification-trained CNN to identify discriminative region localizations in an aerial scene solely by utilizing image labels. Zhang *et al.* [44] proposed a hierarchical weakly supervised learning (HWSL) method to realize pixelwise residential area segmentation by computing the gradient with respect to middle convolutional layers and low-rank matrix recovery [45].

However, there is still a large gap between the power of weakly supervised methods and fully supervised methods in the following two aspects.

1) *Low Accuracy and Poor Boundary Maintenance:* Since pooling operations, such as GMP [37] and global average pooling [38], have been applied in previous methods, it is challenging to generate full-resolution saliency maps, and the boundary maintenance of these methods is unsatisfactory.
2) *Time Consumption:* The gradient maps [36], [39], [40], [44] and the sophisticated postprocessing are widely used in previous weakly supervised methods, which are inevitably time-consuming.

In addition, in practical problems, compared with labeled images, the overwhelming majority of images are unlabeled and directly obtained from sensors without any prior knowledge. This phenomenon is particularly severe in the field of remote sensing, where manual annotations have a high dependence on expert knowledge. Therefore, how to utilize these unlabeled images in a supervised algorithm and let them participate in the training phase presents another challenge.

In this article, we capitalize on the advantages of WSL and fully supervised learning to develop a progressively supervised learning (PSL) method for RSIs. The objective of our proposal is to construct an end-to-end lightweight SOD framework with only imagewise annotation. As shown in Fig. 1, the proposed approach mainly consists of two parts: pseudo saliency map (PSM) generation with auxiliary images and a transition from weakly supervised to fully supervised learning. The key concept is that a feedback saliency analysis network (FSAN) is constructed in a fully supervised manner, where the PSMs generated by WSL are treated as pixelwise ground truth. Subsequently, considering that the PSMs are not accurate enough to train a highly efficient network, we design a denoising loss function to choose the involved pixels, which are more likely to be correct samples. Meanwhile, we introduce curriculum learning to gradually increase the difficulty of the task, thus enhancing the representation capacity.

The major contributions of this article can be summarized as follows.

1) *Pseudo Label Generation With Auxiliary Images:* To make effective use of unlabeled auxiliary images, we provide an appropriate way to generate the PSMs in a weakly supervised manner. Here, only imagewise annotations are required to train a classification network. Then, for auxiliary images, the positive samples determined by the well-trained CNN are fed to generate the PSMs by Grad-CAM. The importation of auxiliary images enhances the number of training samples in the subsequent end-to-end network, thus boosting the performance of the framework to a great extent.
2) *Weakly Supervised to Fully Supervised Learning:* To combine the superiority of fully and WSL, we build an FSAN to regress the prediction of the weakly supervised method (PSMs generated by Grad-CAM). Here, only PSMs are fed to optimize the parameters in training phase. The network is composed of two parts: the construction of the FSAN and curriculum learning in the training phase. In the first part, inspired by the powerful representation ability of feedback connections, we build a deep but lightweight FSAN to avoid overfitting. In the training phase, since the PSMs are not sufficiently accurate, we introduce curriculum learning, which gradually enhances the difficulty by selecting different training pixels. Meanwhile, a new denoising loss function is designed, where general fusion maps (GFMs) are proposed to determine the confidence of each pixel and help select the training samples that are more likely to be correct. The selective strategy favors boundary maintenance and suppressing the background areas.

II. Related Work

In this section, we present a brief review of DL-based fully supervised saliency analysis methods, DL-based weakly supervised methods, and SOD for RSIs.

A. CNN-Based Fully Supervised SOD Methods

Different from the majority of traditional SOD methods, which are based on contrast cues [46], [47], CNN-based

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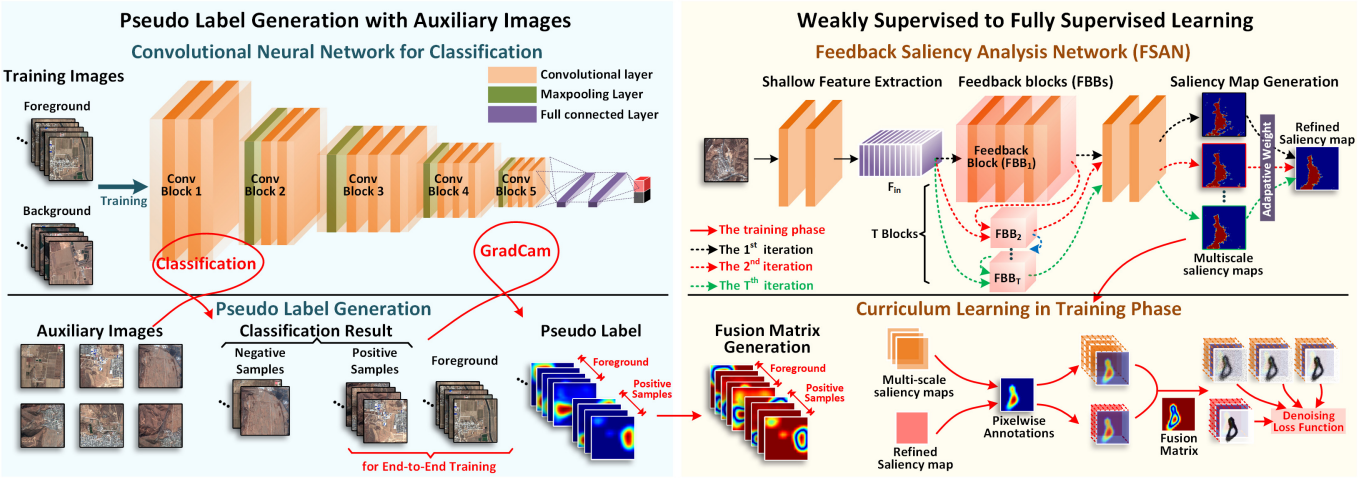


Fig. 1. Flowchart of the proposed PSL method.

fully supervised models contain neurons with variable convolutional layers, and the parameters are trained with pixelwise annotations. As an early attempt, Zhao *et al.* [48] proposed a multicontext DL framework for SOD to solve the problem brought by low-level saliency cues or priors. A global context was constructed to robustly model saliency with few large errors, and a local-context CNN was designed to refine the saliency prediction of the centered superpixel. He *et al.* [49] proposed a superpixelwise CNN approach, where the color uniqueness sequence and color distribution sequence were extracted and fed to the convolutional networks. Wang *et al.* [50] presented a saliency detection algorithm by integrating both local estimation and global search.

All the abovementioned methods were based on fully connected layers, which cannot fully leverage high-level semantic information [3]. Fully convolutional networks [26], which consider pixelwise operations to overcome this problem, have been widely introduced to SOD [30], [51], [52]. Li and Yu [30] proposed an end-to-end deep contrast network that consisted of a pixelwise fully convolutional stream and a segment-level spatial pooling stream. The saliency maps from both streams are fused at the end. Kuen *et al.* [51] proposed a recurrent attentional convolutional–deconvolutional network to detect objects with multiple scales. To solve the instability caused by superpixels, Tang and Wu [52] proposed a saliency detection method by combining region-level saliency estimation and pixelwise saliency prediction with CNNs. Liu *et al.* [53] proposed a pixelwise contextual attention network (PiCANet) to generate an attention map at each pixel over its context region. It only incorporates useful contextual knowledge and depresses other noisy information for each pixel, facilitating saliency prediction.

A deeply supervised net [54], which simultaneously minimizes classification error while making the learning process of hidden layers direct and transparent, has been introduced into SOD and has shown great performance. Liu and Han [55] proposed an end-to-end deep hierarchical saliency network (DHSNet) based on CNNs. It was directly trained using the input images and corresponding ground truth from coarse scale to fine scale and supervised the corresponding learning of each stepwise saliency map. To extract local, global, and contextual information, Tang *et al.* [56] constructed a deeply supervised recurrent CNN that combined the advantages of recurrent CNNs [57] and deeply supervised nets.

B. DL-Based Weakly Supervised SOD Methods

Most of the fully supervised models have been trained with pixelwise annotations; however, these models are very time-consuming and require substantial human resources. Therefore, WSL, which relies on imagewise annotations, has attracted much attention. Considering that only imagewise annotations are available, most previous works have relied on training classification networks to localize objects, which can also be seen as a variant of multiple instance learning [58].

Simonyan *et al.* [36] proposed a weakly supervised object location approach based on the derivative of the class score with respect to the input image, and the location was advanced by GraphCut [59] color segmentation. Oquab *et al.* [37] proposed a WSL method for transferring midlevel image representations and evaluating the output of CNNs on multiple overlapping patches. To minimize the number of parameters while maintaining high performance, Zhou *et al.* [38] built a procedure for generating class activation maps using global average pooling, which can highlight the discriminative object parts detected by the CNN. Then, Selvaraju *et al.* [39] proposed a Grad-CAM approach that used the gradients flowing into the final convolutional layer to produce a coarse localization map. Rebuffi *et al.* [60] conducted a thorough analysis of backpropagation-based saliency methods named NormGrad and proposed a single framework under which several saliency methods can be unified. Zhang *et al.* [61] proposed a weakly supervised SOD model (ScribbleSaliency for short) to learn saliency from scribble labels, where an edge detection network and a gated structure-aware loss are designed to enforce boundaries of the predicted saliency map to align with image edges in the salient regions.

Recently, researchers have focused on a two-step pipeline in semantic segmentation and object location. Pseudo or approximate ground truth is first generated and then used to train a

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segmentation model. Xu *et al.* [62] proposed computing the pseudo ground truth in a natural scene by generating attention maps and class-agnostic saliency maps obtained from an off-the-shelf saliency detector. However, for RSIs, it is difficult to obtain an off-the-shelf detector. Zhang *et al*. [44] proposed a weakly supervised to fully supervised framework for object detection. In contrast with these works, we realize a pixelwise saliency analysis other than bounding-box location results in this article. Cheng *et al.* [63] put forward an effective way to generate high-quality proposals for weakly supervised object detection, where they focus on the quality and the selection of the proposal. Different from their work, we propose to treat the PSM computed by the first step as the input of the supervision of the second stage, in which we introduce a curriculum learning strategy to improve the representation capacity of the framework by progressively increasing the learning difficulty.

C. SOD for RSIs

Saliency analysis has been widely applied for object detection and segmentation in RSIs. Zhang and Yang [18] presented a frequency-domain strategy based on the QFT for salient region detection in RSIs that was time-saving and efficient. Zhang *et al.* [17] proposed an ROI detection algorithm based on multiscale feature fusion for panchromatic RSIs. Zhu *et al.* [19] proposed a target detection method based on both bottom-up (BU) saliency and top-down (TD) saliency. BU saliency uses a graph-based visual saliency model [64] to locate those outstanding areas, and TD saliency uses the near parallelity of line segments. To overcome the hurdle brought by abundant and complex background disturbances, Ma *et al.* [14] proposed an ROI detection method based on superpixel-to-pixel saliency analysis. Zhang *et al.* [9] proposed a global and local saliency analysis (GLSA) model for extracting residential areas in high-spatial-resolution RSIs.

For the past few years, with the development of DL techniques, researchers have come up with many CNN-based methods because of their strong generalization ability. Han *et al.* [65] proposed a practical geospatial object detection framework by combining WSL and high-level feature learning, where the deep Boltzmann machine (DBM) [66] was used due to its feature representation ability. Diao *et al.* [67] proposed an object detection framework that combined the strength of the unsupervised feature learning of deep belief networks (DBNs) [68] and visual saliency. Shao *et al.* [34] proposed a saliency-aware CNN framework for ship detection, comprising comprehensive ship discriminative features such as deep features, saliency maps, and coastline priors. Han *et al.* [65] proposed a weakly supervised object detection algorithm with only binary labels. The method integrates saliency, intra-class compactness, and interclass separability in a Bayesian framework to initialize training examples and iteratively learn the object detector. Yao *et al.* [69] proposed a dynamic curriculum learning strategy that aims to learn the detector by progressively increasing the training data's difficulty with only image-level annotations. To solve the problem that existing weakly supervised object detection methods tend to seek only one object with the highest score, Feng *et al.* [70] proposed an end-to-end progressive contextual instance refinement method to perform weakly supervised object detection in RSIs. In our previous work [13], we proposed a framework that combines multiview learning and attention mechanism (MLAM) for ROI extraction in RSIs.

III. Methodology

In this study, a novel PSL method is not only designed by virtue of low labor costs but can also progress in an end-to-end manner in the testing phase. The images containing salient objects are classified in the foreground category, whereas others are classified in the background category.

There are two types of prior information are used in the whole framework: 1) the imagewise annotations used to train the classification network and 2) the construction of the general saliency map. As shown in Fig. 1, the proposed approach consists of two key steps: 1) PSM generation with auxiliary images and 2) weakly supervised to fully supervised learning. The purpose of the first step is to generate PSMs by Grad-CAM through the well-trained classification CNN. Meanwhile, we propose to generate the PSMs of auxiliary images and feed them into the subsequent FSAN to enhance the diversity of samples. For the second step, we design a new FSAN, where the pseudo annotations (PAs) generated by PSMs are cast as pixelwise ground truth in the training phase. Also, curriculum learning is introduced to make the network suitable for more complicated scenes. Considering that the PSM itself contains some errors, the GFM is computed to reflect the true extent of each pixel. Also, a denoising loss function is designed through the GFM, where the pixels with high confidence are selected to participate in the training phase. Detailed descriptions are presented in the following part.

A. Pseudo-Label Generation With Auxiliary Image

In the field of remote sensing, pixelwise annotations are expensive and time-consuming because expert knowledge is essential, and it is difficult to annotate them by crowdsourcing. In addition, compared with labeled images, there are millions of available unlabeled RSIs that can be easily collected. In this section, we solve these problems in two steps. First, we train a classification CNN with imagewise annotations, which is used to generate PSMs by computing the Grad-CAM of input images. Second, a well-trained CNN is also responsible for positive filtrate samples in auxiliary images. The PSMs of these samples are computed in the same way to enhance the performance of the subsequent network. It should be mentioned that all the operations in part are only required in the training phase.

1) CNN for Classification: As shown in Fig. 2, our classification network is inspired by the component of VGG16 [71], which consists of 13 convolutional layers, each of which is equipped with a rectified linear unit (ReLU) [72] as the activation function. Five max-pooling layers are added to decrease the size of the feature and divide the convolutional layers into five convolutional blocks. Then, two fully connected layers with 256 neurons are added after the convolutional layers to

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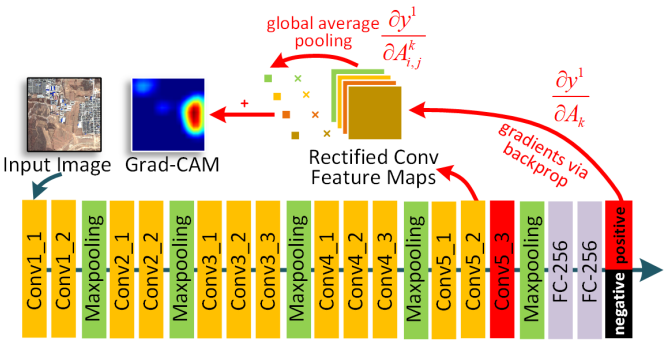


Fig. 2. Architecture of the classification network and the flowchart of Grad-CAM.

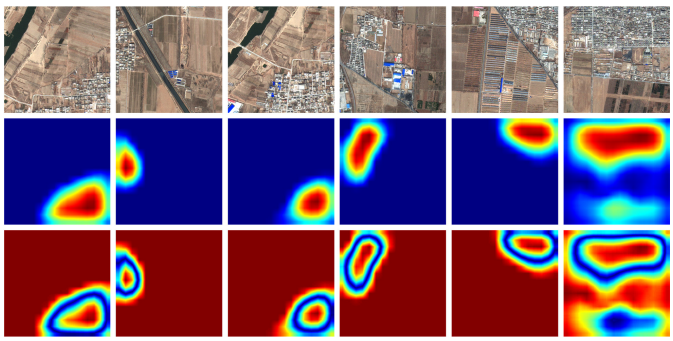


Fig. 3. Visual exhibitions of PSMs and the fusion maps from the training data set (columns 1–3) and auxiliary images (columns 4–6) from the GeoEye-1 data set. Each row from top to bottom is original RSIs, PSMs, and fusion maps.

enhance the nonlinear ability of the framework. Since there are two categories in the training data set, we set two neurons followed by the softmax activation function to compute the probability of the corresponding classes. The weights of 13 convolutional layers are initialized from VGG16 [71], which has been pretrained over the ImageNet data set, and the weights of fully connected layers are randomly initialized.

In detail, the classification CNN is a mapping function from the $\mathbb{R}^{m\times n\times h}$ image space to the $\mathbb{R}^{1\times 2}$ decision space. The training data set is denoted as $\mathbf{I}=\{I_1,\ldots,I_M\}$, $I_i\in\mathbb{R}^{m\times n\times h}$. Given the training image I_i, the corresponding annotation is denoted as $y_i\in\mathbb{R}^{1\times 2}$, $y_{ij}\in\{0,1\}$ denotes the probability of the ith image belonging to the jth category and satisfies

$$\sum_j y_{ij}=1. \tag{1}$$

In this article, $y_i=[0,1]$ means that I_i belongs to the foreground category, whereas $y_i=[1,0]$ means that I_i belongs to the background category.

The class score of the ith sample, denoted as $f(I_i;\theta)^j$, is the output of the jth neuron before the softmax activation function. The probability can be computed as

$$S_{ij}=\frac{\exp\left(f(I_i;\theta)^j\right)}{\sum_{c=1,2}\exp\left(f(I_i;\theta)^c\right)}. \tag{2}$$

In this article, the training procedure generally follows Krizhevsky *et al.* [24] with a cross-entropy loss function:

$$L(I_i,y_i;\theta)=-\sum_{j=1}^{2}y_{ij}\cdot\ln\left(S_{ij}\right) \tag{3}$$

$$L_{\text{classification}}(\theta)=\sum_{I_i\in\mathbf{I}}L(I_i,y_i;\theta) \tag{4}$$

where $\theta=\{W,b\}$ is the combination of weights W and biases b. The optimization procedure can be formulated as

$$\theta^*=\arg\min_\theta\left\{-\sum_{I_i\in\mathbf{I}}\sum_{j=1,2}y_{ij}\cdot\ln\left(S_{ij}\right)\right\} \tag{5}$$

where the well-trained parameter is denoted as θ^*.

2) Pseudo-Label Generation: The purpose of this section is to generate the PSMs of training images and auxiliary images. Inspired by Selvaraju *et al.* [39], Grad-CAM is selected as the baseline to generate pixelwise saliency maps.

For the training images with imagewise annotations, only foreground images are utilized in the following, and the background images are abandoned. Given an image I, the class score $f(I;\theta^*)^j$ is the output of the jth neuron before the softmax activation function. $j=1$ represents the foreground category, and $j=2$ represents the background category. Since salient objects are the foreground category, only the output of the first neuron is taken into consideration when generating Grad-CAM.

As suggested in [39], the gradient information flowing into the last convolutional layer (Conv5_3 in Fig. 2) is used to understand the importance of each neuron for a decision of interest. The kth feature map in the last convolutional layer is denoted as A^k. We first compute the class score for class 1 with respect to feature maps A^k as $\frac{\partial f(I;\theta^*)^1}{\partial A^k}$. These gradients flowing back are used to generate the importance weight by global average pooling

$$\alpha_k=\frac{1}{Z}\sum_s\sum_t\frac{\partial f(I;\theta^*)^1}{\partial A^k_{s,t}} \tag{6}$$

where $A^k_{s,t}$ is the intensity at position (s,t) in the kth feature map.

The Grad-CAM of training images, which are also referred to PSMs, is computed as

$$S_{\text{Grad-CAM}}=\text{Relu}\left(\sum_k\alpha_k^cA^k\right). \tag{7}$$

For auxiliary images, because the data set is a mixture of foreground and background images, we first utilize the well-trained CNN to divide them into two parts: positive samples that are likely to be in a foreground category and negative samples that are likely to be in a background category. For the training data set, only the positive samples are used to generate the Grad-CAM by (6) and (7).

Figs. 3 and 4 show the visual exhibitions of PSMs and auxiliary images from the GeoEye-1 and SPOT5 data sets. It can be seen that the PSMs can roughly highlight the salient objects (residential areas). Moreover, it can also exclude many background areas. However, since there are four max-pooling layers in front of the Conv5_3 layer in VGG16, it is unable to generate a full-resolution saliency map.

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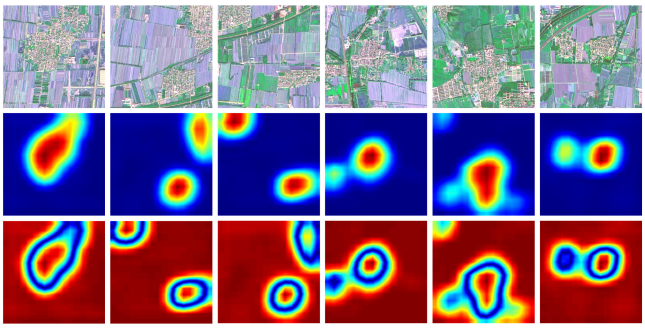


Fig. 4. Visual exhibitions of PSMs and the fusion maps from the training data set (columns 1–3) and auxiliary images (columns 4–6) from the SPOT5 data set. Each row from top to bottom is original RSIs, PSMs, and fusion maps.

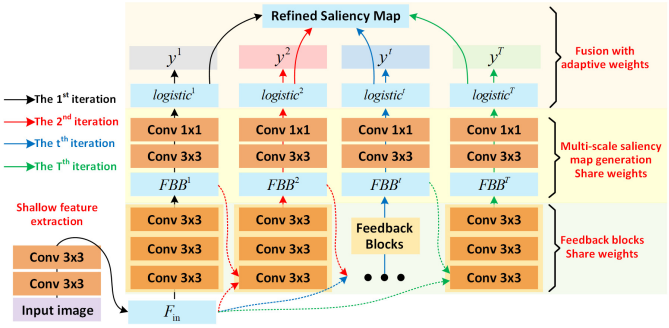


Fig. 5. Network architecture of the FSAN.

B. Weakly Supervised to Fully Supervised Learning

In this section, several modules are designed to enhance the accuracy of the saliency prediction, including the construction of FSAN, the design of denoising loss function, and the introduction of curriculum learning.

Here, we build an FSAN trained by curriculum learning, where the binary segmentation of the PSMs is regarded as pixelwise PA. It is worth mentioning that the images with true saliency labels are not involved in the training phase. In our proposal, the PSM is treated as a transition between WSL and fully supervised learning. If we simply train the network under the PAs without any constraint, it will be unable to provide a satisfactory extraction result since the PAs are not accurate. To solve this problem, the fusion binary map is generated by the PSMs, which can reflect the confidence of each pixel. The main principle during the generation process is that the pixels with extremely high and low saliency values are very likely to be correct and thus provide a large contribution to the loss function. In contrast, the pixels in the middle range are of high uncertainty and thus should provide a smaller contribution to the loss function.

In Section III-B1, we first present the network architecture of the FSAN. Then, in Section III-B2 we provide a detailed description of the curriculum learning in the training phase.

1) Feedback Saliency Analysis Network: For an input image x, the corresponding PSM is denoted as $S_{\text{Grad-CAM}}$. The FSAN consists of three parts: 1) shallow feature extraction; 2) feedback blocks; and 3) saliency map generation. The structure of the proposed FSAN is shown in Fig. 5.

In the first part, for an input image x, two convolutional layers are employed to extract shallow features as follows:

$$F_{\text{in}} = \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(x)) \tag{8}$$

where $\text{Conv}_{s\times s}(\bullet)$ is an $s \times s$ convolutional layer with m channels. If there are no special instructions, each convolutional layer is followed by a ReLU activation function.

Inspired by the powerful representation ability of feedback connections [73], we design a new feedback block for saliency analysis with varying levels of texture. Given that there are T iterations in the whole network, the feedback block at the tth iteration receives the hidden state from the previous iteration F_{FBB}^{t-1} through a feedback connection and shallow features F_{in}. The feedforward propagation of the FBB is denoted as $f_{\text{FBB}}(\bullet)$, and each FBB shares the same weighting parameters. The output of the tth iteration is computed as

$$F_{\text{FBB}}^{t} = f_{\text{FBB}}\left(\left[F_{\text{FBB}}^{t-1}, F_{\text{in}}\right]\right) \tag{9}$$

where $[F_{\text{FBB}}^{t-1}, F_{\text{in}}]$ denotes the concatenate of F_{FBB}^{t-1} and F_{in}. For the first iteration, we tile F_{in} two times to ensure that the input dimension of FBB is consistent. Specifically, the $f_{\text{FBB}}(\bullet)$ operation is the stack of three convolutional layers with a kernel size of 3×3, where

$$f_{\text{FBB}}(\cdot) = \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(\text{Conv}_{3\times3}(\cdot))). \tag{10}$$

For the saliency map generation part, the convolutional layers in each iteration share the same weighting parameters. There are two layers in this part: the first layer is a 3×3 convolutional layer with m channels, and the second layer is a 1×1 convolutional layer with two channels, which represent the logistic outputs of the foreground and background categories. The activation function of the last convolutional layer is a softmax to regular range of the output saliency maps to [0, 1]

$$\text{logistic}^{t} = \left(\text{Conv}_{1\times1}\left(\text{Conv}_{3\times3}\left(F_{\text{FBB}}^{t}\right)\right)\right) \tag{11}$$

$$y^{t} = \text{soft}\max\left(\text{logistic}^{t}\right). \tag{12}$$

For a given image x, the output of the FSAN is a $T + 1$ channel saliency map. The first T saliency maps are the output of T iterations, and the last saliency map is the fusion of them by adaptive weights. Here, adaptive weights $[\alpha^1, \ldots, \alpha^T]$ are also determined in the training phase. The refined saliency map is denoted as y^{T+1}

$$y^{T+1} = \text{soft}\max\left(\sum_{t=1}^{T}\left(\alpha^{t} \cdot \text{logistic}^{t}\right)\right). \tag{13}$$

Besides, since the pooling layer or upsampling layers are not used in the proposed FSAN, our work can provide a full-resolution saliency prediction, thus maintaining the spatial information as much as possible.

2) Curriculum Learning in Training Phase: Curriculum learning [74], which gradually increases the difficulty of the task, has shown great power to enhance the representation capacity in the training procedure. In our work, curriculum learning is introduced by selecting different pixels in different iterations according to the intensity of the PSMs. As shown in Fig. 3 (line 2) and Fig. 4 (line 2), the pixels with large

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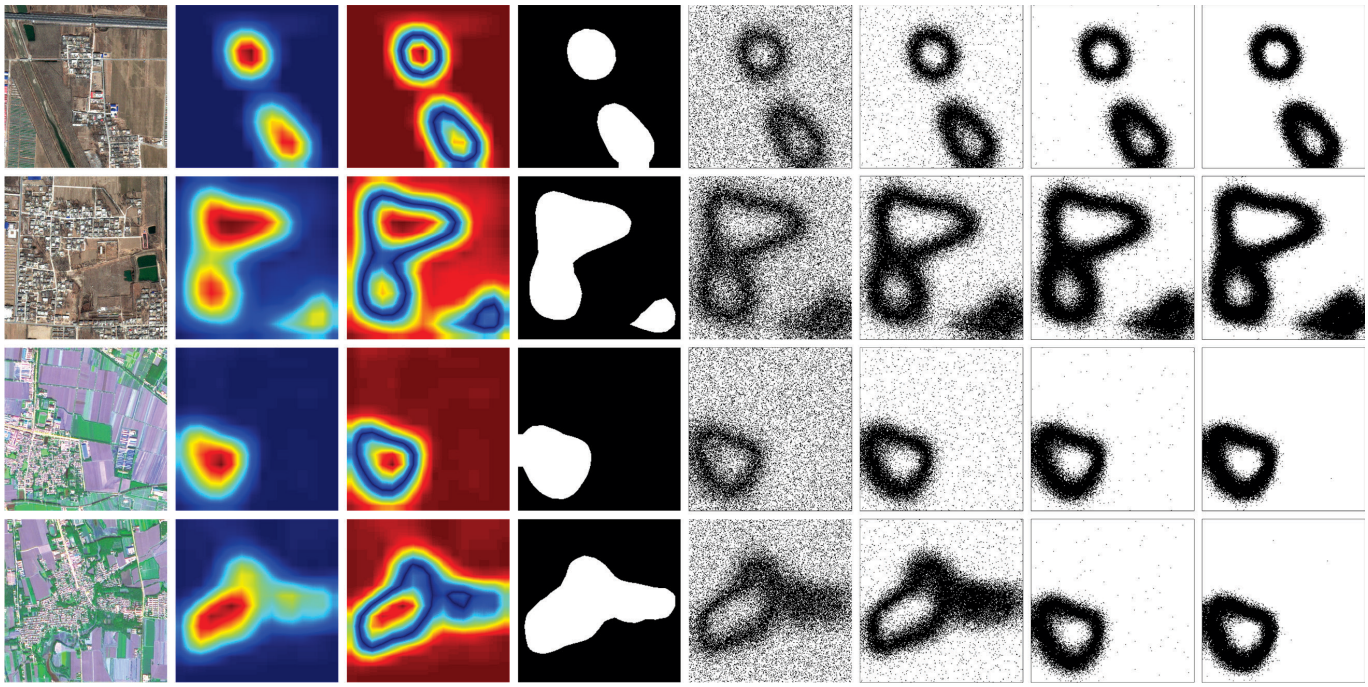


Fig. 6. Visual exhibitions of RSIs from the GeoEye-1 data set (the first row) and SPOT-5 data set (the second row). Each column from left to right is (a) original RSIs, (b) PSM, (c) GFMs, (d) PAs, (e) binary fusion map of the first iteration, (f) binary fusion map of the second iteration, (g) binary fusion map of the third iteration, and (h) binary fusion map of the fourth iteration.

intensities are very likely to be the salient objects, and thus, they are more likely to have a correct decision. Similarly, pixels with small intensities are more likely to be background areas and have a great chance to be a true judgment. In other words, the pixels distributed over the middle range are likely to be a false judgment. For a given image x, considering that the PSMs are scaled into the range of [0, 1], we propose to compute the GFM Prob, which measures the possibility of correct judgment, as follows:

$$\text{Prob} = 2 \cdot |S_{\text{Grad-CAM}} - 0.5|. \tag{14}$$

The last column in Figs. 3 and 4 shows the GFM. It can be observed that the areas with higher response (in red) are more likely to be correct judgment, whereas the areas with lower response (in blue) are more likely to be incorrect judgment.

As shown in Fig. 7, for different iterations, we propose to design the loss function with various difficulties according to the principle of curriculum learning. Specifically, for a given image x, the output of the FSAN can be written as $[y,^1\ y,^2 \ldots, y^T,\ y^{T+1}]$, where y^{T+1} is the weighted average of $y,^1\ y,^2 \ldots, y^T$. The loss function at the tth iteration is designed as

$$\text{loss}_t = -\sum_{i=1}^{|x|} \left(\text{Prob}_i^t \left[G_i \cdot \log\left(P\left(y_i^t = 1\right)\right) + (1 - G_i) \cdot \log\left(P\left(y_i^t = 0\right)\right)\right]\right) \tag{15}$$

where Prob^t is the fusion binary map in the tth iteration and Prob_i^t is the intensity of the ith pixel, which can only be 0 or 1. G_i is the PAs for the ith pixel, which are Otsu binary segmentations from PSMs. For each iteration, Prob^t is regarded as a selection of pixels in the input images. For instance, if $\text{Prob}_i^t = 0$, the ith pixel is unable to participate in the training phase. We provide the principle of how to generate the fusion binary map Prob^t in the tth iteration as follows.

We enforce the curriculum to learn multiple cases by adding stochastic disturbance into the GFM. The main rule is that, for shallow iterations, more training samples are selected, while for deep iterations, only the pixels with large confidence are selected. Thus, with increasing iteration, the difficulty of the task becomes large. Therefore, the results should be more accurate for larger iterations. Thus, we add Gaussian noise with $\mu = 0$ and $\sigma^2 = 0.5/t$ for the tth iteration to the GFM Prob. Then, we obtain the fusion binary map Prob^t under the threshold $= 0.5$. Fig. 6 shows the visual exhibition of the GFMs and the fusion binary maps when $T = 4$. It shows that with the decreasement of the variance, less uncertain pixels are selected in the training phase.

The loss function of the whole network is computed as

$$L = \sum_{t=1}^{T+1} \text{loss}_t. \tag{16}$$

The training phase can be summarized in Table I.

C. Implementation Details

The PC was equipped with an Intel Core i7-9700 K CPU with 3.6 GHz × 8, 32 GB of memory, and GPU NVIDIA Geforce RTX 2080Ti. We provide the implementation details of the classification CNN and the FSAN as follows.

For the classification CNN, the original image is cropped to 256 × 256 × 3 with imagewise annotations, which reflect whether the input image contains salient objects. We train the model with a cross-entropy loss function. Then, we use

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TABLE I
ALGORITHM: TRAINING PROCEDURE OF THE FSAN

Input: training image and auxiliary images dataset $\boldsymbol{X}$.
For x in $\boldsymbol{X}$ **do**:
If: x belongs to foreground training images **do**:
Compute the pseudo saliency maps $S_{Grad\text{-}CAM}$ by Grad-CAM as equation (6) and (7).
Add x and $S_{Grad\text{-}CAM}$ into the training dataset $\boldsymbol{X'}$ and PSM dataset $\boldsymbol{S}$ of subsequent network.
If: x belongs to auxiliary images **do**:
Decide the category of x.
If x is a positive sample **do**:
Compute the pseudo saliency maps $S_{Grad\text{-}CAM}$ by Grad-CAM as equation (6) and (7).
Add x and $S_{Grad\text{-}CAM}$ into the training dataset $\boldsymbol{X'}$ and PSM dataset $\boldsymbol{S}$ of subsequent network.
For x in $\boldsymbol{X'}$ **do**:
Compute the general fusion map *Prob* as equation (14).
Compute the pixelwise annotations with Otsu threshold.
For t in T **do**:
Generate the fusion binary map by adding Gaussian noise with $\mu = 0$ and $\sigma^2 = 0.5/t$, and then segment it with threshold=0.5.
Compute the loss function in t^{th} iteration as equation (15).
Compute the final loss function as equation (16).
Compute the loss function of the whole dataset as: $Loss = \sum_{x \in X'} L$.
Train the FSAN with Adam optimization method.
Output: The well-trained FSAN, which can detect the salient object in an end-to-end manner.

the stochastic gradient descent algorithm with a momentum of 0.9 and a weight decay of 0.0005 to optimize the whole model. We train the classification CNN with 500 iterations with batch size = 10 at the learning rate of 10^{-3}.

For the FSAN, the PAs are treated as pixelwise ground truth in the training phase. The size of the input images is 256 × 256 × 3. We apply the Adam [75] optimizer with $\beta_1 = 0.9$ to train the networks. Considering the complexity of the task, the FSAN is trained with 2000 iterations with batch size = 8 at the learning rate of 10^{-3}.

In the testing phase, only feedback saliency analysis is used to generate the pixelwise results in an end-to-end manner. The refined saliency map $y^{T|+|1}$ is computed as the final saliency map. It is worth mentioning that our proposal can accept images of arbitrary size due to their fully convolutional structures.

IV. EXPERIMENTS

A. Data Sets and Methods for Comparison

We evaluate the proposed PSL framework on two remote sensing data sets, including the GeoEye-1 data set and SPOT5 data set. GeoEye-1 satellite images have an original resolution of 0.46 m, and the input image is composed of bands 3, 2, and 1. SPOT5 satellite images have an original resolution of 2.5 m, and the input image is composed of bands 2, 1, and 3. For both data sets, the raw images are 768 × 768 × 3 pixels.

TABLE II
CONFIGURATION OF TWO DATA SETS DERIVED FROM GEOEYE-1 AND SPOT-5 SATELLITES

Dataset	Training images		Auxiliary images	Testing images
	Foreground	Background		
GeoEye-1 dataset	129	151	416	10
SPOT5 dataset	79	81	623	12

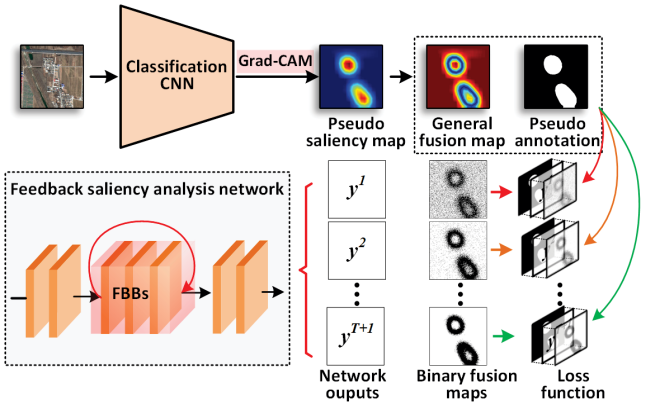


Fig. 7. Curriculum learning in the training phase.

We first resize them to 256 × 256 × 3 before feeding them to our approach.

Residential areas, which contain useful information, are regarded as salient objects in our research. To train the classification network, images, including residential areas, are foreground samples, whereas others are background samples. Table II shows the configuration of the data set. It can be observed that the number of auxiliary images is much greater than the number of training images.

B. Results and Comparisons

In this section, we compare our proposed PSL with 11 state-of-the-art methods. Among them, Itti [76], DSR [77], and RBD [78] are traditional methods used for natural scenes, MFF [17], GLSA [9], HWSL [44], and MLAM [13] are used for RSIs, and Grad-CAM [39], Grad-CAM++ [40], NormGrad [60], ScribbleSaliency [61], MLAM [13], and HWSL [44] are weakly supervised methods with imagewise annotations. In Section IV-B1, the saliency maps and the SOD results are compared by visual comparison and quantitative analysis.

1) Visual Comparison: Figs. 8–10 show the visual comparisons of the saliency maps and the extraction results for the GeoEye-1 and SPOT5 data sets. Fig. 11 shows the zoomed-in view comparison for these two data sets. It can be observed that traditional methods cannot provide a satisfactory result. Itti *et al.* [76] generated the features by a set of linear "center-surround" operations akin to visual receptive fields, so the pixels at the center have a high response. However, since the distribution of residential areas is random, the "center-surround" assumption is not appropriate for RSIs. In work [77], the image boundaries are first extracted via superpixels, so when residential

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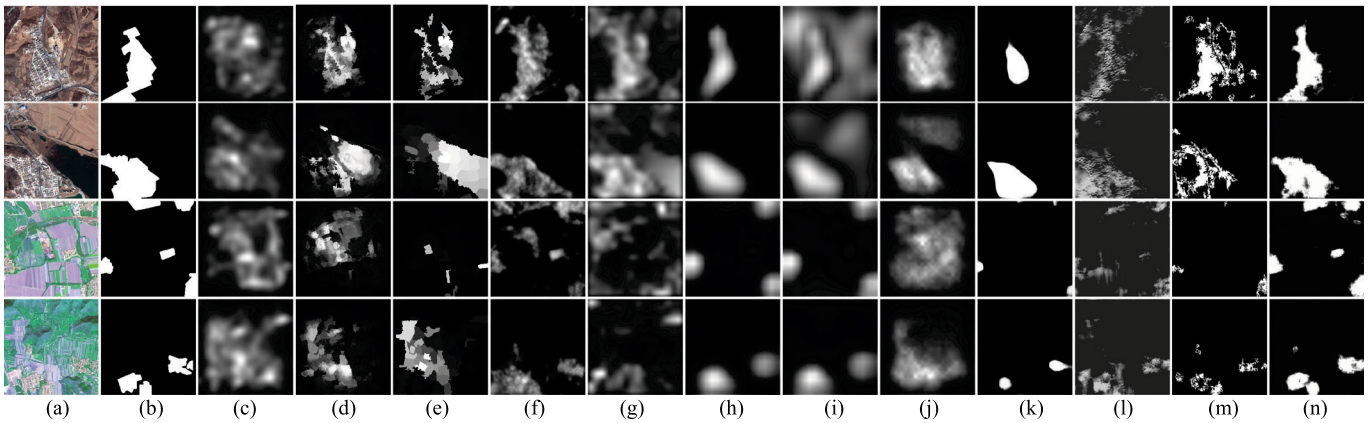


Fig. 8. Visual comparison of saliency maps for: 1) GeoEye-1 data set (the first two rows) and 2) SPOT5 data set (the last two rows). From left to right are (a) input images, (b) ground truth, (c) Itti, (d) DSR, (e) RDB, (f) MFF, (g) GLSA, (h) Grad-CAM, (i) Grad-CAM++, (j) NormGrad, (k) ScribbleSaliency, (l) HWSL, (m) MLAM, and (n) our proposal.

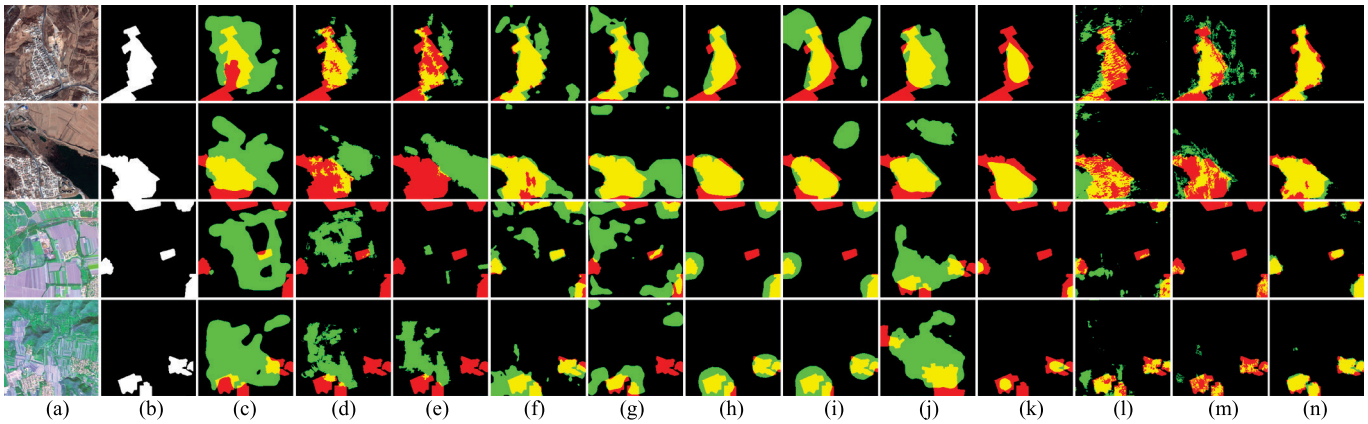


Fig. 9. Visual comparison of extraction results for: 1) GeoEye-1 data set (the first two rows) and 2) SPOT5 data set (the last two rows). From left to right are (a) input images, (b) ground truth, (c) Itti, (d) DSR, (e) RDB, (f) MFF, (g) GLSA, (h) Grad-CAM, (i) Grad-CAM++, (j) NormGrad, (k) ScribbleSaliency, (l) HWSL, (m) MLAM, and (n) our proposal. Yellow: true positive, green: false positive, red: false negative, and black: true negative.

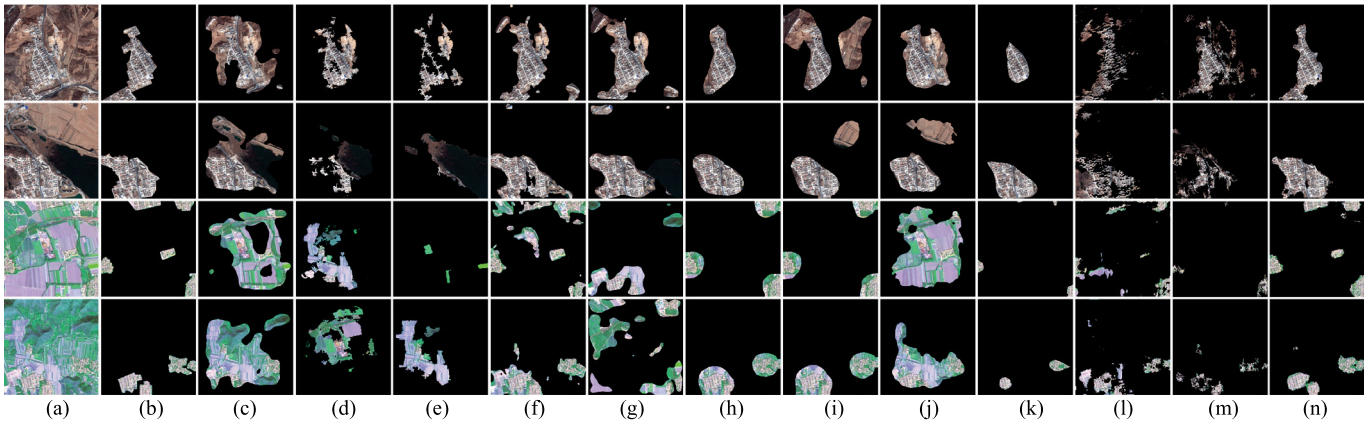


Fig. 10. Visual comparison of extraction results for: 1) GeoEye-1 data set (the first two rows) and 2) SPOT5 data set (the last two rows). From left to right are (a) input images, (b) ground truth, (c) Itti, (d) DSR, (e) RDB, (f) MFF, (g) GLSA, (h) Grad-CAM, (i) Grad-CAM++, (j) NormGrad, (k) ScribbleSaliency, (l) HWSL, (m) MLAM, and (n) our proposal.

areas appear near the boundaries, poor results (as shown in Figs. 9 and 10, column (c), last two rows). RBD [78] cannot obtain satisfactory performance because they consider that object regions are much less connected to image boundaries than background regions. MFF [17] and GLSA [9], which are specifically designed for RSIs, can provide better results for the GeoEye-1 data set. However, when facing a more complex scene, such as the SPOT5 data set, they do not work very well.

For weakly supervised methods, since Grad-CAM [39] and Grad-CAM++ [40] are computed by the last convolutional layers, they cannot generate a saliency map with full-resolution and clear boundaries. As suggested in [60], ResNet [25] is selected as the baseline of NormGrad [60] and layers 2–4 are selected as target layers. It can be observed that the extraction results of NormGrad [60] sometimes are not so satisfying, especially in the SPOT5 data set, since the sharper grayscale

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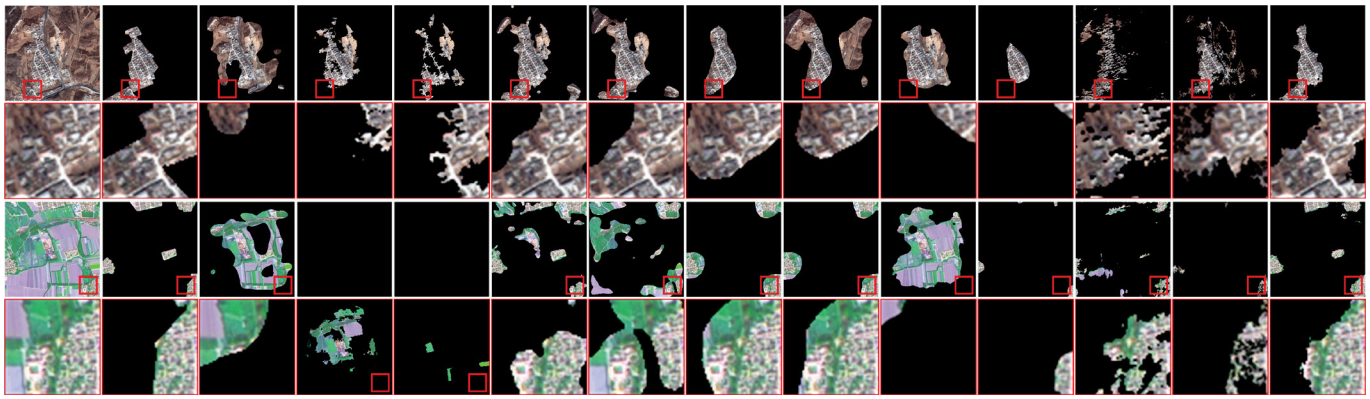


Fig. 11. Local sight of extraction results for: 1) GeoEye-1 data set (the first two rows) and 2) SPOT5 data set (the last two rows). From left to right are (a) input images, (b) ground truth, (c) Itti, (d) DSR, (e) RDB, (f) MFF, (g) GLSA, (h) Grad-CAM, (i) Grad-CAM++, (j) NormGrad, (k) ScribbleSaliency, (l) HWSL, (m) MLAM, and (n) our proposal.

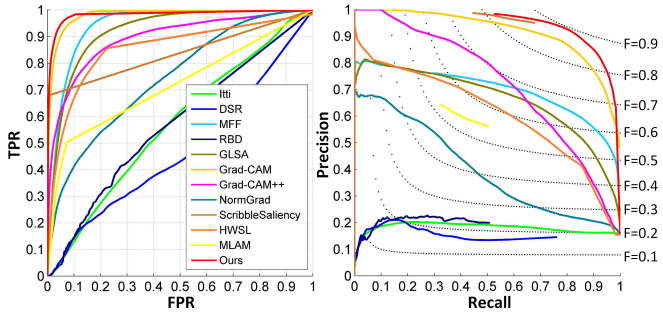


Fig. 12. Evaluation metrics of the saliency maps for the GeoEye-1 data set: (a) ROC curve and (b) PR curve.

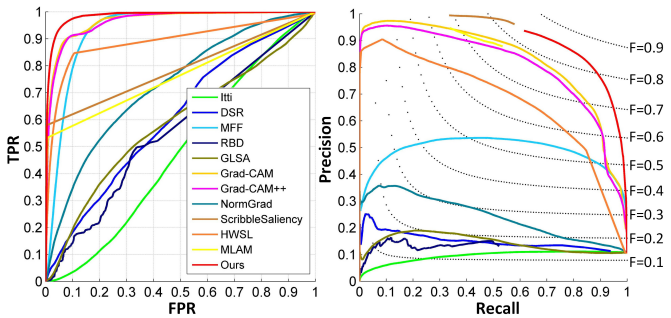


Fig. 13. Evaluation metrics of the saliency maps for the SPOT5 data set: (a) ROC curve and (b) PR curve.

changes in RSI are very sensitive to the gradients. As for ScribbleSaliency [61], the supervision of this method is much stronger than our proposal. It uses scribble annotations in the training phase, which includes three categories of supervision signal: foreground, background, and unknown pixels. Thus, we relabel GeoEye-1 and SPOT5 data set with scribble annotations to run this method. As shown in Fig. 9 [column (k)], ScribbleSaliency [61] can achieve very high true positive rates and the false-negative rate is relatively high. We consider that it is a common shortage for scribble annotations in RSIs, where the partial cross-entropy loss takes only scribble annotation as supervision and the weak correlation between pixels further increases the difficulties. For HWSL [44], since it integrates all gradient maps of each convolutional layer, the saliency maps are very sensitive to small changes. MLAM [13], which combines MLAM, can provide a better boundary maintenance than other methods. However, since the attention mechanism relies heavily on the threshold when extracting the color and texture features, the extraction result is not stable and always influenced by the contribution of pixels in the whole images. Compared with other saliency analysis methods, the proposed PSL method can compute a full-resolution saliency map with clear boundaries and consistent saliency values.

2) Quantitative Analysis: We evaluate our proposal and other methods in terms of a receiver operator characteristic (ROC) curve, area under the curve (AUC), precision–recall (PR) curve, precision, recall, and F-measure.

For a given saliency map, a binary map is obtained by segmenting it with varying quantization thresholds and then compared with the ground truth to compute the false positive rate (FPR), true positive rate (TPR), precision, and recall for an image. FPR and TPR are then depicted in the ROC curve, and the precisions and recalls are depicted in the PR curve. We use the F-measure to evaluate the extraction results. We first segment the binary map by Otsu thresholding. For each binary map, we compute the precision, recall, and F-measure as follows:

$$\text{Precison} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

$$F_\beta = \frac{(1+\beta^2)\text{Precision} \times \text{Recall}}{\beta^2(\text{Precison} + \text{Recall})} \tag{19}$$

where TP is the number of true positives, FP is the number of false positives, and FN is the number of false negatives. Similar to [79], β^2 is set to 0.3 in our work. Figs. 12 and 13 show the comparison of the ROC curves and PR curves. The quantitative results of the GeoEye-1 and SPOT5 data sets are reported in Table III. The best results are marked in bolds.

CNN-based methods, including Grad-CAM [39], Grad-CAM++ [40], NormGrad [60], ScribbleSaliency [61], MLAM [13], HWSL [44], and the proposed PSL, show better

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TABLE III
OBJECTIVE METRICS VALUES OF THE GEOEYE-1 DATA SETS

	GeoEye-1				SPOT5			
Metrics	AUC	Precision	Recall	F-measure	AUC	Precision	Recall	F-measure
Itti	0.5719	0.1813	0.3791	0.2038	0.4818	0.0868	0.3139	0.1019
DSR	0.4700	0.2017	0.1820	0.1857	0.6079	0.1658	0.3099	0.1788
RDB	0.5684	0.2350	0.1092	0.1758	0.5590	0.0966	0.2296	0.1064
MFF	0.9443	0.5748	0.8890	0.6199	0.9318	0.4125	**0.9221**	0.4682
GLSA	0.9236	0.4488	**0.9206**	0.5028	0.5850	0.1533	0.3772	0.1749
Grad-CAM	0.9817	0.8450	0.8361	0.8396	0.9671	0.6488	0.8552	0.6829
Gram-CAM++	0.8954	0.5738	0.7916	0.5902	0.9649	0.6103	0.8647	0.6503
Norm-Grad	0.7449	0.3173	0.5311	0.3455	0.7262	0.2059	0.5948	0.2386
Scribble-Saliency	0.8378	**0.9761**	0.5345	0.7982	0.7887	**0.9895**	0.4132	0.6810
HWSL	0.8510	0.5749	0.6451	0.5786	0.8839	0.6929	0.6087	0.6596
MLAM	0.7190	0.5832	0.3975	0.5184	0.7638	0.9164	0.4274	0.6976
Ours	**0.9825**	0.9409	0.7747	**0.8941**	**0.9837**	0.8560	0.8261	**0.8448**

TABLE IV
TIME CONSUMPTION OF WEAKLY SUPERVISED METHODS

Methods	Grad-CAM	Gram-CAM++	NormGrad	ScribbleSaliency	HWSL	MLAM	Ours
Running Time s	0.1966s	0.2304s	1.2328s	0.0738s	13.1260s	4.1549s	**0.0167**

performances than traditional methods in both data sets because the task-specific features extracted by CNN are very helpful for saliency analysis. Among the CNN-based methods, our proposal can achieve a higher TPR with the same FPR in the ROC curve for the GeoEye-1 data set and SPOT5 data set; thus, the AUC is higher than other methods. For the PR curve, our method achieves consistent and favorable performance against the state-of-the-art methods in both data sets. As for precision and recall values, although the recall values of the GLSA [9], Grad-CAM [39], and Grad-CAM++ [40] are higher than our proposed method, their precision is inevitably low because they judge many pixels as residential areas. Besides, we find that although the precision of ScribbleSaliency [61] is higher than our proposal, the recall is much lower. The main reason is that the scribble annotations only provide part of the information about the training images, and the partial loss function lets the model tend to recognize pixels inside the salient object, while the pixels around the boundary are very hard to identify. Overall, our proposal achieves the highest F-measure among all comparative methods in both data sets.

Table IV shows the average time of processing one image for different weakly supervised methods. Here, we evaluate all methods ten times. In each time, we choose ten images for testing. We find that since Grad-CAM [39] needs to compute the gradient maps and Grad-CAM++ [40] needs to use the Laplace operator, the running time of them is much longer than proposal. As for HWSL [44], NormGrad [60], and MLAM [13] because these methods have to compute the gradient maps several times, the running time of these methods are all longer than 1 s for one image with size $256 \times 256 \times 3$. ScribbleSaliency [61] utilizes VGG16 [71] for feature extraction, so the size of the network is larger than our proposal, thus leading to longer running time. For the proposed method, the number of parameters and the layers are decreased by the feedback blocks, so it can realize end-to-end SOD and the speed increase by ten times compared with Grad-CAM [39] and Grad-CAM++ [40].

V. DISCUSSION

In this section, we discuss the impact of each part, including the auxiliary images, the denoising loss function, the curriculum learning, the refinement operation, and the iterations, to show the rationale of the proposed PSL framework. It can be observed that the class distribution of the residential areas and the backgrounds is unbalanced. As suggested in [80], PR curves provide a more accurate picture of an algorithm's performance when dealing with a highly skewed data set. Therefore, only the PR curve and F-measure are used in this section. We first design seven comparison experiments that indicate the importance of each part via an ablation study.

A. Configuration of Ablation Study

1) The Influence of the Depth: We construct three FSANs with $T = 1$, 2, and 3 to evaluate whether the depth of iteration has an influence on the performance. In the following experiments, T is set to three consistently.

2) W/o Auxiliary Images: We only use the PSMs of training images with imagewise labels to train the FSAN.

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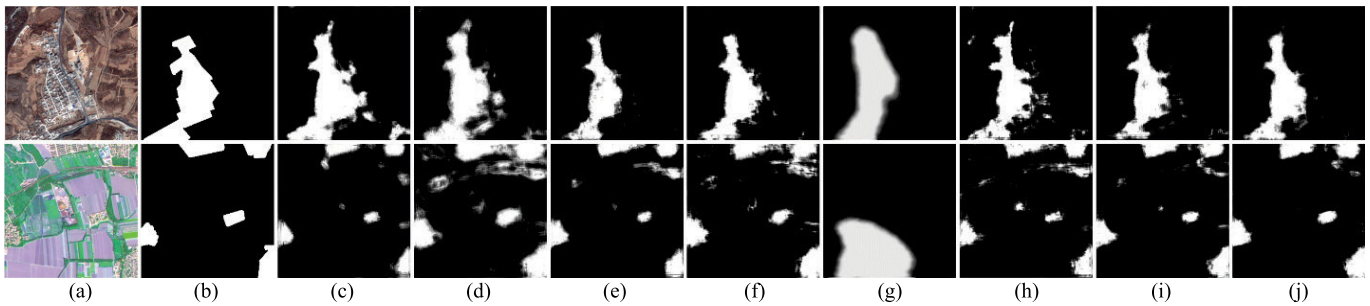


Fig. 14. Visual comparison of saliency analysis in the ablation study for: 1) GeoEye-1 data set (the first rows) and 2) SPOT5 data set (the second rows). From left to right are (a) input images, (b) ground truth, (c) w/o auxiliary images, (d) w/o denoising loss, (e) w/o curriculum learning, (f) w/o curriculum refine, (g) with Unet, (h) ours ($T = 1$), (i) ours ($T = 2$), and (j) ours ($T = 3$).

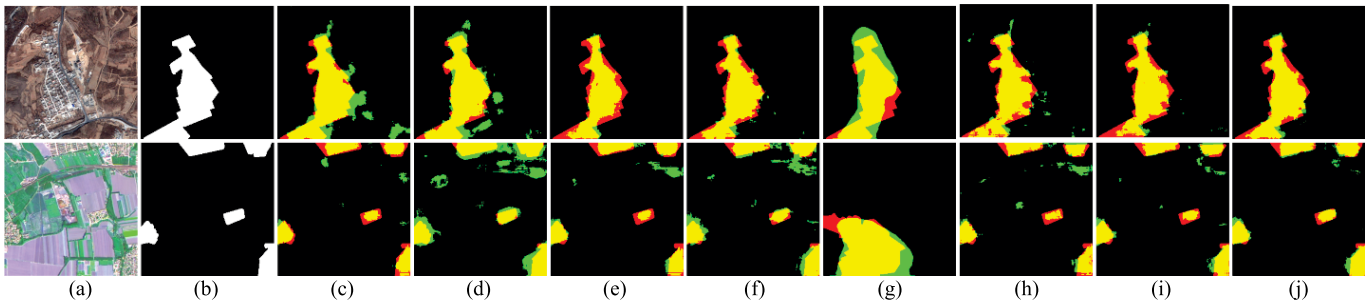


Fig. 15. Visual comparison of extraction results in the ablation study for: 1) GeoEye-1 data set (the first rows) and 2) SPOT5 data set (the second rows). From left to right are (a) input images, (b) ground truth, (c) w/o auxiliary images, (d) w/o denoising loss, (e) w/o curriculum learning, (f) w/o curriculum refine, (g) with Unet, (h) ours ($T = 1$), (i) ours ($T = 2$), and (j) ours ($T = 3$).

3) W/o Denoising Loss: We train the FSAN with the traditional cross-entropy loss function without fusion binary maps as

$$\text{loss}_t = -\sum_{i=1}^{|x|}\left(G_i \cdot \log\left(P\left(y_i^t = 1\right)\right) + (1 - G_i) \cdot \log\left(P\left(y_i^t = 0\right)\right)\right). \tag{20}$$

In this experiment, the FSAN is used to directly regress the prediction of the weakly supervised results.

4) W/o Curriculum Learning: We use the PSM instead of the fusion binary map in each iteration in the loss function, so the pixels involved in the training phase are always the same during all iterations. The loss function is as follows:

$$\text{loss}_t = -\sum_{i=1}^{|x|}\Big(S_{\text{Grad-CAM}} \cdot \left[G_i \cdot \log\left(P\left(y_i^t = 1\right)\right) + (1 - G_i) \cdot \log\left(P\left(y_i^t = 0\right)\right)\right]\Big). \tag{21}$$

5) W/o Refine: We train a network without a refining operation in (13), and the output of the last iteration y^T is regarded as the final output.

6) With Unet: We use Unet rather than the proposed FSAN to learn the mapping between the input images and the PSMs. It should be mentioned that we use the same loss function to train these two networks.

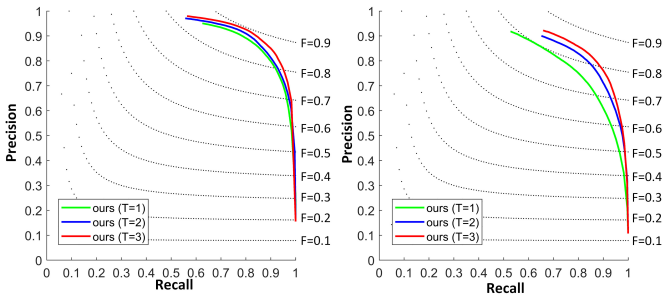


Fig. 16. PR curve of the saliency maps by our proposal with $T = 1.2.3$ for (a) GeoEye-1 data set and (b) SPOT5 data set.

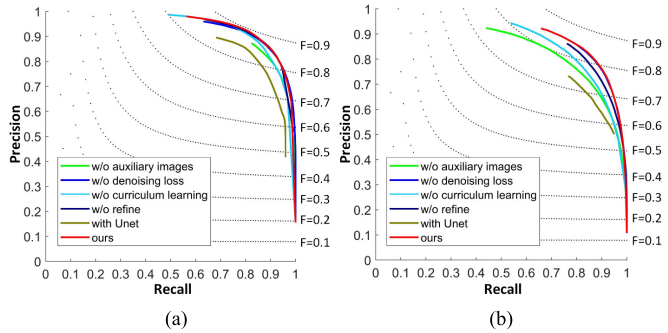


Fig. 17. PR curve of the saliency maps in the ablation study for (a) GeoEye-1 data set and (b) SPOT5 data set.

B. Ablation Study

1) The Influence of the Depth: Figs. 14 and 15 [columns (h)–(j)] show the generated saliency maps and the detection results with $T = 1$, 2, and 3, respectively. It can be observed that with the increasing number of iterations, there will be less noise in the saliency maps. Fig. 16 and Table V report the PR curve and F-measure generated by our proposal with $T = 1$, 2, and 3. It can be observed that the framework with deeper layers can achieve a higher PR curve and F-measure.

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TABLE V
F-MEASURE WITH T = 1.2.3 OF THE GEOEYE-1 AND SPOT5 DATA SETS

T	dataset	Precision	Recall	F-measure
1	GeoEye-1	0.9089	**0.7770**	0.8733
2		0.9316	0.7638	0.8847
3		**0.9409**	0.7747	**0.8941**
1	SPOT5	**0.8708**	0.6613	0.8020
2		0.8475	0.7824	0.8260
3		0.8560	**0.8261**	**0.8448**

TABLE VI
F-MEASURE IN ABLATION STUDY OF THE GEOEYE-1 AND SPOT5 DATA SETS

Dataset	Method	Precision	Recall	F-Measure
GeoEye-1	w/o auxiliary images	0.7776	**0.9260**	0.8053
	w/o denoising loss	0.8449	0.9100	0.8575
	w/o curriculum learning	**0.9559**	0.7135	0.8838
	w/o refine	0.9318	0.7813	0.8906
	with Unet	0.7672	0.8887	0.7836
	Ours	0.9409	0.7747	**0.8941**
SPOT5	w/o auxiliary images	**0.8898**	0.6773	0.8170
	w/o denoising loss	0.6389	**0.9513**	0.6892
	w/o curriculum learning	0.8856	0.7055	0.8266
	w/o refine	0.7290	0.8771	0.7565
	with Unet	0.6218	0.8798	0.6589
	Ours	0.8560	0.8261	**0.8448**

2) The Contribution of the Auxiliary Images: Figs. 14 and 15 [column (c)] show the results generated without auxiliary images. For the GeoEye-1 data set, the utilization of auxiliary images can dramatically decrease the number of false positives. As shown in Fig. 17, for the SPOT5 data set, our proposal can enhance the recall, which means that there are fewer false negatives. Since the various samples are fed to the FSAN in our proposal, the introduction of auxiliary images can boost the F-measure for both data sets.

3) The Contribution of the Denoising Loss Function: Figs. 14 and 15 [column (d)] show the results generated without the traditional loss function. We find that there are many nonresidential areas with high responses in the saliency maps. The reason is that since the PSMs generated by Grad-CAM are insufficiently accurate, the network cannot be trained well only with inaccurate ground truth. In particular, the extraction results without denoising loss always have higher recall and lower precision compared with our proposal because the PSM has the same properties, as shown in Tables III and VI. The experiment indicates that training with the denoising loss function can obtain better performance for both data sets.

4) The Contribution of the Curriculum Learning: Figs. 14 and 15 [column (d)] show the results generated without curriculum learning. As shown in Fig. 17, compared with the framework without curriculum learning, we can obtain more true positives; thus, the recalls in both data sets are much higher than those without curriculum learning. The main reason is that, in our proposal, random Gaussian noise is added to the GFMs, which increases the diversity of the samples and can improve the efficiency of the network.

5) The Contribution of the Feedback Structure and Refine Operation: Figs. 14 and 15 [column (g)] show the saliency analysis results and the extraction results with Unet, respectively. It can be observed that the ROC and PR curve of our proposal is higher than Unet, which shows the superiority of the FSAN. Figs. 14 and 15 [column (f)] show the saliency analysis results and the extraction results without refining, respectively. We find that the detection results of w/o refine contain many false positive judgments because only the features under a single scale are used to generate the saliency maps. Since the low-level features and high-level features are integrated with PSL, we can obtain better boundary maintenance, and there are fewer false-positive judgments.

VI. CONCLUSION

In this article, we proposed a novel SOD method based on PSL for RSIs. First, we propose a pseudo-label generation method to generate PSMs for training samples and auxiliary images in a weakly supervised manner. Then, to improve computational efficiency, we construct an FSAN. Finally, inspired by curriculum learning, we design a new denoising loss function to further reduce the effect brought by missing judgment in PSMs and enhance the detection accuracy. In Section V, we demonstrated the impact of each part through an ablation study. In summary, we establish a PSL method that can achieve high-accuracy and lightweight SOD for RSIs. In the future, we intend to further explore PSL for multiple class SOD in RSIs to further reduce the dependence on human labor.

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